

ECON 675 Capstone Final Report

What Exactly Does Gold Protect You From?

A Gold vs. Silver Test Across Equities, and Bonds

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1. Executive Summary

This paper investigates what gold and silver actually protect investors from, using a portfolio-based analysis across multiple crises and regions. We assemble weekly data on equities, long-term government bonds, and precious metals for the United States, Europe, Japan, and the United Kingdom, and evaluate gold and silver within a mean–variance framework. Rather than relying only on correlations, we compare assets and portfolios using drawdowns, volatility, Sharpe ratios, and crisis-specific performance during the dot-com bust, the Global Financial Crisis, the European sovereign-debt crisis, and the COVID-19 shock.

The results show that gold behaves as a state-dependent safe haven, while silver largely does not. In calm markets, optimal minimum-variance and tangency portfolios allocate primarily to high-Sharpe equity indices (especially NASDAQ), with little or no weight on gold or silver. During severe crises, however, the optimizer rotates sharply into gold-heavy portfolios, substantially reducing maximum drawdowns and stabilizing cumulative wealth relative to equity-only, silver-only, and traditional 60/40 stock–bond benchmarks. Silver, by contrast, remains volatile, often sells off with equities in stress, and is rarely selected in risk-minimizing or maximum-Sharpe portfolios.

We also compare static and dynamic strategies. Portfolios that re-estimate covariances and expected returns over rolling windows and adjust weights accordingly consistently outperform static minimum-variance, equal-weight, and 60/40 allocations, both in terms of Sharpe ratios and drawdown control. These findings generalize across regions: efficient frontiers in Europe, Japan, and the United Kingdom all assign a central role to gold in minimum-variance portfolios, with silver appearing mainly in high-risk, high-return corners of the frontier. Overall, the analysis shows that gold protects investors primarily from large, equity-driven crashes and global “risk-off” episodes, while silver behaves more like a cyclical, high-beta asset rather than a reliable safe haven.

2. Introduction

In periods of market stress, investors increasingly look for assets that can protect wealth when traditional portfolios are under pressure. Gold, in particular, has long been marketed as a “safe haven” that holds or even gains value when equities sell off, interest rate expectations shift abruptly, or the macroeconomic outlook deteriorates. Silver is often discussed in the same breath as gold, yet it is simultaneously an industrial metal whose cash flows are more tightly linked to the business cycle. These competing narratives raise a core question: what exactly do gold and silver actually protect investors from, and how robust is that protection across crises, regions, and portfolio constructions?

A large empirical literature has evaluated whether gold behaves as a safe haven for stocks and, in some cases, bonds. Many of these papers focus on one benchmark equity index, a single

crisis episode, or a specific country, most commonly the United States (e.g., Baur & Lucey, 2010; Baur & McDermott, 2010). Far fewer studies treat silver as a potential safe haven in its own right, and even fewer directly compare gold and silver in a unified portfolio framework that spans multiple crises, asset classes, and regions. As a result, it remains unclear whether gold's perceived safe-haven status reflects a broad, robust phenomenon or a more narrow pattern concentrated in certain episodes, markets, or portfolio designs. Silver's role is even less well understood: it could either behave as "poor man's gold" or as a high-beta, pro-cyclical asset.

This paper addresses that gap by taking a portfolio-based view of gold and silver across major global crises and regions. Using weekly data on U.S. equities, long-term Treasury yields, and precious metal prices, together with regional equity indices for Europe and Japan, we evaluate how gold and silver behave during the dot-com bust, the Global Financial Crisis, the European sovereign debt crisis, and the COVID-19 shock, as well as during surrounding non-crisis periods. Figures 1–3 document the behavior of U.S. equities, gold, and silver over time, including total S&P 500 returns, drawdowns, and normalized price levels with U.S. Treasury yields. Figures 4–6 extend this perspective globally by plotting long-term yields and equity indices for Europe and Japan alongside metals. Figures 7–8 then zoom in on drawdowns and volatility in the United States, overlaying gold and silver on S&P 500 risk metrics to visually assess whether the metals move more like hedges or like amplifiers.

Beyond descriptive evidence, we embed gold and silver directly in a mean–variance portfolio framework. Tables 1 and 2 report U.S. summary statistics and full-sample minimum-variance and tangency portfolio weights when investors can allocate across equities, bonds, gold, and silver. Tables 3A and 3B contrast crisis and non-crisis portfolio weights and performance, clarifying whether gold and silver only help during crashes or also improve long-run risk–return tradeoffs. Table 4 extends the portfolio analysis across regions, while Tables 5 and 6 summarize U.S. performance metrics and test whether dynamic, crisis-conditional allocations to gold and silver materially outperform static buy-and-hold strategies. Taken together, these tables and figures allow us to evaluate safe-haven claims in terms of drawdowns, volatility, Sharpe ratios, and maximum losses, rather than relying solely on simple correlations.

Our main findings are that gold behaves much closer to a safe haven than silver, but its protective value is highly state-dependent. Gold tends to reduce portfolio drawdowns and improve risk-adjusted performance during the most acute phases of the Global Financial Crisis and the COVID-19 crash, especially in U.S. equity–bond portfolios. Outside these episodes, however, the benefits of large, static gold allocations are more modest, and in some specifications they come at the cost of lower long-run returns. Silver, by contrast, typically resembles a high-beta asset: it occasionally offers short-lived diversification benefits but generally does not deliver the same crisis protection as gold. These patterns largely persist when we extend the analysis to European and Japanese equity markets, though the strength of the safe-haven relationship varies across regions and crises.

The remainder of the paper is organized as follows. The next section briefly reviews the related literature on safe-haven assets, precious metals, and crisis performance. We then describe the data and crisis definitions, outline our empirical framework for measuring returns, drawdowns, and volatility, and for constructing optimal portfolios with and without precious metals. The subsequent section presents our main results for the United States and then for Europe and Japan, highlighting how the figures and tables in the Appendix map onto each piece of our analysis. We conclude by summarizing what gold and silver actually protect investors from, discussing the limitations of our study, and drawing out implications for portfolio construction and risk management.

3. Literature Review

A large empirical literature evaluates whether gold behaves as a hedge or a safe haven relative to traditional financial assets. Baur and Lucey (2010) provide one of the foundational definitions: a hedge is an asset that is uncorrelated or negatively correlated with stocks or bonds on average, while a safe haven is an asset that is uncorrelated or negatively correlated specifically in periods of market stress. Using this framework for the United States, the United Kingdom, and Germany, they find that gold is typically a hedge and, in extreme equity-market downturns, a safe haven, though the safe-haven property tends to be short-lived. Baur and McDermott (2010) extend the analysis internationally and show that gold functions as both a hedge and a safe haven for major European stock markets and the U.S., but offers weaker protection for markets such as Japan, Canada, Australia, and large emerging economies. These early studies highlight two key themes: gold's protective role is state-dependent, and its effectiveness varies across regions.

Subsequent work refines this view by focusing on volatility regimes and alternative benchmarks. Hood and Malik (2013) compare gold with the VIX index and conclude that, over their sample, VIX serves as a stronger hedge and safe haven for U.S. equities than gold, and that gold is at best a weak safe haven under changing stock-market volatility. Other studies examine whether gold can hedge not only equity risk, but also exchange-rate or macro-financial risks more broadly, often finding that gold's hedging ability is asymmetric and concentrated in severe downturns rather than normal times. More recent contributions also explore gold's behavior in the context of the global financial cycle, finding that its safe-haven properties can depend on the nature and source of the underlying shock. Overall, this strand of the literature suggests that gold is rarely an "always-on" safe haven; instead, its usefulness depends on both the market and the type of turmoil considered.

A second strand of research broadens the focus to other precious metals, including silver. Lucey and Li (2015) examine the time-varying safe-haven properties of gold, silver, platinum, and palladium for U.S. stock and bond markets. They document that, in some periods, silver, platinum, and palladium act as safe havens when gold does not, implying that no single metal consistently dominates. Li (2017) extends this analysis to eleven countries and finds that all four metals can, at times, serve as safe havens against equities and bonds, with the identity of the

“best” safe haven varying by country, asset class, and episode. Lahiani, Mefteh-Wali, and Vasbieva (2021) focus on the COVID-19 period and show that several precious metals exhibit safe-haven behavior vis-à-vis the S&P 500, though the strength and persistence of protection differs across metals and between short- and long-run horizons.

More recent work has begun to look specifically at gold and silver together. Caporale and Gil-Alana (2023) analyze the long-run linkages between gold, silver, and a wide set of stock indices using fractional integration and cointegration methods. They find that, over 2010–2019, gold and silver often behave as long-run diversifiers rather than strict safe havens, but during the COVID-19 period both metals show stronger evidence of safe-haven behavior as their price differentials relative to equities become less mean-reverting. Other studies examining precious metals in China and other markets similarly report that gold and silver can act as hedges or safe havens in some states of the world, but these roles are episodic and sensitive to model specification. Across this literature, silver is typically more volatile and more closely tied to industrial activity than gold, which makes its safe-haven role more fragile and less consistent.

A related line of research investigates safe-haven behavior during specific crises. Several papers show that gold provided some protection during the Global Financial Crisis and, to a lesser extent, during earlier episodes such as the dot-com bust, but that the magnitude of protection differs across markets and metrics (e.g., correlations vs. tail co-movements vs. drawdowns). During the COVID-19 shock, the literature is more mixed: some studies find that gold retained its safe-haven properties relative to equities, while others argue that its behavior was closer to that of a diversifier, especially once central-bank interventions stabilized markets. A growing set of papers also compares gold with newer candidates such as Bitcoin, typically finding that Bitcoin’s safe-haven properties are weaker and more unstable, while gold remains the more reliable crisis asset. These crisis-focused results reinforce the notion that safe-haven status must be evaluated episode by episode rather than assumed to be permanent.

Although many of these studies include some form of portfolio analysis, the treatment is often limited. Baur and Lucey (2010) and Hood and Malik (2013) both embed metals in simple stock–bond–gold portfolios and show that gold can modestly improve risk-adjusted performance in certain states, but they typically focus on one country and a single benchmark equity index. More recent work evaluates “false” safe havens by testing whether candidate assets actually improve out-of-sample portfolio performance under realistic constraints, finding that some assets that appear to be safe havens in correlation-based tests deliver disappointing portfolio outcomes (Kaczmarek et al., 2022). However, there remains relatively little research that systematically compares gold and silver within a unified mean–variance portfolio framework across multiple crises, regions, and asset classes (equities, bonds, and foreign exchange), and that moves beyond static buy-and-hold allocations to consider dynamic, crisis-conditional strategies.

Taken together, the literature establishes three key points that motivate this paper. First, gold often behaves as a hedge and sometimes as a safe haven, but the strength and duration of that protection are state dependent and vary across countries. Second, silver and other precious metals can also display safe-haven behavior, yet their performance is more episodic and is influenced by their industrial uses and higher volatility. Third, most existing studies are either

single-country or single-market in focus, emphasize correlations or tail co-movements rather than explicit portfolio performance, and rarely provide a direct, side-by-side comparison of gold and silver across multiple crises and regions. This paper contributes to the literature by placing gold and silver in the same mean–variance portfolio setting across four major crises (dot-com, Global Financial Crisis, European debt crisis, and COVID-19) and multiple regions, and by evaluating their roles not only through correlations but also through drawdowns, volatility, Sharpe ratios, and the performance of both static and dynamic portfolio strategies, as summarized in the tables and figures reported in our Appendix.

4. Data and Methodology

4.1 Asset universe and sample period

Our empirical work is built around a set of liquid, representative assets that allow us to compare gold and silver with traditional portfolio building blocks across multiple regions. For the United States, we use:

- *Equities*: S&P 500 Index (SPX), treated as a broad, market-cap–weighted proxy for U.S. equities.
- *Sovereign bonds*: 10-year and 30-year U.S. Treasury yields, which we use to construct long-duration bond returns.
- *Precious metals*: Spot prices for gold and silver in U.S. dollars.
- *Cash proxy*: A short-term risk-free rate used to compute excess returns and Sharpe ratios.

For our international analysis, we add equity indices and government bond yields for Europe, Japan, and the United Kingdom, together with gold and silver series expressed in local-currency terms or in U.S. dollars where local data are not available over the full period. These regional series allow us to test whether gold and silver behave similarly as safe havens outside the United States and to construct region-specific efficient frontiers and optimal portfolios.

Our primary sample period runs from the mid-1990s through early 2025, subject to data availability for each asset and region. U.S. series for equities, bonds, and metals are available back to 1995, while consistent regional data for Europe and Japan begin around 2000, and for the United Kingdom around 2004. These staggered start dates match the coverage shown in the Appendix: U.S. series are plotted from 1995 onward (e.g., Figures 1–3, 7–8, 11–12, 16–25), while the regional plots and frontiers begin in 2000 or 2004 (Figures 4–6, 9–10, 13–15, 26–30).

All series are sampled at a weekly frequency using end-of-week observations. We use weekly data for two reasons. First, it smooths out some of the noise in daily returns while still capturing the dynamics of crises and recoveries. Second, weekly observations provide a long effective sample for estimating covariance matrices and rolling portfolio weights without relying on very short windows. Figures 1–6 in the Appendix provide a visual overview of the underlying data, including U.S. annual equity returns and drawdowns, normalized U.S. prices and yields, global sovereign yields, and regional equity and metal price indices.

4.2 Crisis and control windows

To examine safe-haven behavior, we explicitly partition the sample into **crisis** and **control** windows, rather than relying only on full-sample averages. Following standard dating conventions and our own event-study design, we define four major crisis episodes:

1. *Dot-Com Bust (2000–2002)*
2. *Global Financial Crisis (2007–2009)*
3. *European Sovereign Debt Crisis (2010–2012)*
4. *COVID-19 Crisis (2020–2021)*

Weeks that fall outside these intervals are classified as control periods, with particular attention to relatively tranquil subperiods such as 2003–2006, 2013–2019, and the post-COVID recovery. This partition allows us to compare how gold and silver behave during extreme stress versus more normal conditions.

Crisis windows are used in two ways. First, we use them to annotate and interpret the time-series figures—for example, when reading U.S. drawdowns and volatility in Figures 2, 3, 7, and 8. Second, we use them to compute crisis vs. control performance measures for individual assets and portfolios. In particular, the U.S. portfolio Sharpe ratios and returns by crisis and control (Figures 20–21) and the crisis-specific portfolio weights and performance summaries (Figures 23, 25; Tables 3A–3B) are all based on this crisis classification.

4.3 Return construction and risk measures

For all price indices and metal prices, we construct weekly log returns as

$$r_t = \ln \left(\frac{P_t}{P_{t-1}} \right),$$

where P_t is the end-of-week price level. For sovereign bonds, we use weekly changes in long-term yields, along with approximate duration-based price relationships, to obtain weekly bond returns consistent with a long-duration Treasury exposure. Risk-free returns are proxied using a short-term interest rate series. Excess returns are defined as asset returns minus the risk-free rate and are used in all Sharpe ratio, tangency portfolio, and efficient frontier calculations.

To characterize safe-haven properties, we employ several complementary risk measures:

- *Drawdowns*: For each asset or portfolio, the drawdown at time t is defined as the percentage decline from the previous peak:

$$D_t = \frac{P_t - \max_{s \leq t} P_s}{\max_{s \leq t} P_s}.$$

Figure 2 shows U.S. S&P 500 drawdowns with gold and silver overlays, while Figure 7 displays rolling 12-month drawdowns for the S&P 500 with metal overlays, highlighting whether gold and silver tend to cushion or exacerbate equity losses.

- *Volatility*: We compute annualized volatility as

$$\sigma_{\text{ann}} = \sqrt{52} \cdot \text{Std}(r_t),$$

using weekly returns. Figure 8 compares the volatility of gold and silver with that of the S&P 500 over time, and Figures 19–21 report volatility-based Sharpe ratios for key portfolios at the full-sample and crisis vs. control horizons.

- *Correlations and relative volatility*: While the Appendix focuses more on visual and portfolio-based diagnostics, our normalized price plots (Figures 3, 5, 6, 9, and 10) and volatility comparisons (Figure 8) implicitly convey cross-asset co-movements and risk ratios, especially between metals and equities across regions.

These measures allow us to move beyond static correlations and examine whether gold and silver behave like hedges or safe havens in terms of actual losses, volatility, and portfolio-level risk.

4.4 Mean–variance portfolio framework

To evaluate the portfolio role of gold and silver, we embed all assets in a standard mean–variance framework. Let μ denote the vector of average weekly excess returns and Σ the covariance matrix of weekly excess returns for the asset set under consideration (e.g., U.S. equities, bonds, gold, and silver). For each specification, we consider two benchmark portfolios:

1. **Minimum-variance portfolio (MVP):**

$$\min_w w^\top \Sigma w \quad \text{subject to} \quad \mathbf{1}^\top w = 1,$$

where w is the vector of portfolio weights, and $\mathbf{1}$ is a vector of ones. This portfolio focuses purely on risk reduction.

2. **Tangency (maximum-Sharpe) portfolio:**

$$\max_w \frac{w^\top \mu}{\sqrt{w^\top \Sigma w}} \quad \text{subject to} \quad \mathbf{1}^\top w = 1.$$

This portfolio trades off risk and return, using the risk-free rate as the benchmark.

Unless otherwise stated, portfolio weights are allowed to vary freely but must sum to one; we do not impose leverage or short-sale constraints in the core analysis, though the resulting optimal weights remain economically plausible. Table 1 reports basic U.S. summary statistics (means, volatilities, and correlations) that feed into this framework. Table 2 then presents the full-sample U.S. MVP and tangency portfolio weights, highlighting the extent to which gold and silver enter optimal portfolios when we ignore crisis/control distinctions.

4.5 Rolling and dynamic portfolios

A key innovation of this paper is to contrast static, full-sample portfolios with dynamic, rolling portfolios that update their weights as information accumulates. We implement two rolling schemes:

- A 52-week rolling window, which we use to estimate minimum-variance portfolio weights over time for the United States (Figure 11).

- An 8-week rolling window with smoothing, which we use to dampen the week-to-week noise in estimated weights and focus on medium-term shifts in risk and correlation patterns. Figures 12–15 show smoothed 8-week rolling MVP weights for the United States, Europe, Japan, and the United Kingdom, respectively.

These rolling portfolios are then compared with static benchmarks in terms of cumulative wealth, drawdowns, and Sharpe ratios:

- *Cumulative wealth and drawdowns:* Figure 16 plots the cumulative wealth trajectories of key U.S. portfolios (e.g., SPX-only, stock–bond, stock–bond–gold, stock–bond–gold–silver), while Figure 17 compares the drawdowns of the minimum-variance portfolio with those of the S&P 500. Figure 18 contrasts rolling versus static portfolio wealth paths to show whether re-optimizing in real time adds value or simply increases turnover and estimation error.
- *Performance metrics:* Figures 19–21 report portfolio Sharpe ratios and annualized returns for the full sample and for crisis versus control windows, and Table 5 provides a consolidated set of U.S. performance measures (returns, volatilities, Sharpe ratios, and maximum drawdowns) for the main portfolios. Table 6 summarizes robustness checks, including dynamic versus static portfolios and alternative weighting schemes, and shows how sensitive our conclusions are to the choice of strategy.

Dynamic portfolios are also analyzed at the regional level. Figure 26 presents regional efficient frontiers using median points for Europe, Japan, and the United Kingdom, while Figure 27 compares Sharpe ratios for dynamic versus static portfolios across regions. Figures 28–30 then provide region-specific frontiers (Japan, United Kingdom, and Europe, respectively), focusing on the combination of minimum variance and maximum return in each market. These regional results build directly on the rolling weight paths in Figures 13–15 and on the regional portfolio weights reported in Table 4.

4.6 Crisis-specific and regional efficient frontiers

To connect safe-haven behavior more tightly to crises, we also construct crisis-specific efficient frontiers and portfolio weights. In the U.S. case, we:

- Re-estimate mean returns and covariances separately for crisis and control windows.

- Compute crisis-only and control-only MVPs and tangency portfolios.
- Compare the resulting weights and performance metrics across states.

Figure 22 shows the resulting optimal U.S. portfolio weights based on weekly data for the full sample, while Figure 23 displays minimum-variance weights in crisis versus control periods, making visible how the model shifts toward or away from gold and silver during stress. Figure 24 plots the U.S. efficient frontier (weekly returns, annualized) for the full sample, and Figure 25 shows crisis-specific minimum-variance portfolio weights, which allow us to see whether the safe-haven role of gold is strongest in particular episodes such as the Global Financial Crisis or COVID-19.

At the regional level, we follow the same logic. Using regional equity indices, local bond yields, and gold and silver prices, we compute regional MVPs and tangency portfolios and trace out efficient frontiers. Table 4 summarizes regional portfolio weights, while Figures 26–30 show regional frontiers and the dynamic vs. static performance of these portfolios. This regional perspective ensures that our conclusions about gold and silver are not driven solely by the U.S. and that we can test whether metals behave differently in Europe, Japan, and the United Kingdom.

5. Results

5.1 Baseline Asset Behavior and Crisis Episodes in the United States

We begin by characterizing the behavior of U.S. equities and precious metals over the full sample. Figure 1 shows S&P 500 total annual returns, highlighting deep negative years in the early 2000s, the Global Financial Crisis (GFC), and the COVID-19 shock. These episodes anchor our crisis windows and illustrate the magnitude of equity drawdowns that any safe-haven asset must mitigate. In parallel, Figure 2 overlays indexed gold and silver paths on SPX drawdowns; gold typically peaks when equity losses are most severe, while silver's response is more erratic and often coincides with equity distress rather than offsetting it.

Figure 3 provides a joint view of normalized gold, silver, equity, and Treasury yield series for the United States (2000–2025). Gold trends upward over the long run with comparatively smooth compounding, whereas silver displays pronounced boom-bust cycles, especially around 2011–2013 and during COVID-19. Equities exhibit high long-run growth punctuated by sharp collapses, while long-term Treasuries show declining yields and episodic price rallies in crises. Together, Figures 1–3 show that gold behaves like a crisis-sensitive store of value in the U.S. sample, whereas silver remains more cyclical and speculative.

Table 1 summarizes full-sample annualized returns, volatility, and Sharpe ratios for the U.S. asset set. Gold earns a modest mean return (0.92% per year) with relatively low volatility (7.50%) and a Sharpe ratio of 0.128, while silver earns a slightly higher mean return (0.99%) but almost double the volatility (13.31%) and a much lower Sharpe ratio of 0.077. Equities dominate in risk-adjusted terms—NASDAQ in particular—with a Sharpe ratio of 0.244 versus 0.200 for the S&P 500 and negative Sharpe ratios for both 10- and 30-year Treasuries. This confirms that gold is not a return-maximizing asset but offers comparatively efficient risk compensation relative to silver within the U.S. universe.

Figure 7 refines this picture by plotting rolling 12-month S&P 500 drawdowns with metal overlays. Gold systematically experiences shallower and shorter drawdowns than equities during major stress episodes, while silver’s drawdowns often mirror the equity pattern. This visual evidence supports the view that gold helps cushion cumulative losses precisely when they matter most, whereas silver does not reliably preserve capital at crisis lows.

5.2 Global Interest Rates, Regional Metals, and Equity Divergence

Extending beyond the U.S., Figure 4 plots 10-year and 30-year government bond yields across all regions from 1995 to 2025. Yields trend downward over the sample, with large downward spikes during the GFC and COVID-19, capturing the global “flight-to-safety” into sovereign bonds. This declining yield environment compresses potential bond returns and helps explain why Treasuries contribute negatively to Sharpe ratios in Table 1.

Figures 5 and 6 move to the cross-market and cross-region perspective. Figure 5 shows normalized gold and silver prices in Europe and Japan; gold rises persistently in both regions, particularly around the GFC and COVID-19, while silver again displays more cyclical and crisis-sensitive swings. Figure 6 plots normalized equity indices globally (1995 = 100) and reveals significant divergence across regions—U.S. and some European equities outperform over the long run, while other indices lag or experience prolonged stagnation. These patterns emphasize that any global safe-haven claim must hold across heterogeneous equity paths and interest-rate regimes.

Figures 9 and 10 explicitly compare normalized gold and silver prices across all regions. Gold appreciates steadily in U.S., European, and Japanese terms, especially during global stress periods, underscoring its role as a broadly recognized crisis asset. Silver rises strongly in some windows but exhibits large reversals across regions, confirming its higher sensitivity to local industrial cycles and risk-on environments. Together, Figures 4–6, 9, and 10 show that the global macro backdrop—falling yields and divergent equity performance—accentuates the difference between gold’s stable crisis demand and silver’s cyclical behavior.

5.3 Volatility, Drawdowns, and Portfolio Wealth in the United States

Figure 8 directly compares the rolling realized volatility of gold and silver to S&P 500 volatility. Gold’s volatility ratio is typically below 1 in normal times—indicating that gold is less volatile than the equity benchmark—but approaches or slightly exceeds 1 during severe crises. Silver’s volatility ratio is almost always above 1, with pronounced spikes in 2008, the European debt crisis, and COVID-19. This confirms that gold tends to dampen portfolio volatility, whereas silver amplifies it.

We then evaluate how these differences translate into portfolio outcomes. Figures 16–18 trace the cumulative wealth and drawdowns of key portfolios. Figure 16 shows that a minimum-variance (MinVar) portfolio and a gold-heavy allocation compound more smoothly than equity-only or silver-heavy portfolios; the equity-only and silver-only strategies experience deeper and more prolonged drawdowns. Figure 17 makes this explicit by comparing drawdowns for the MinVar portfolio and the S&P 500: the MinVar strategy consistently limits peak-to-trough losses, particularly during the GFC and COVID-19. Figure 18 contrasts rolling versus static portfolio wealth paths, with rolling optimization delivering higher terminal wealth and less severe drawdowns than static counterparts, indicating that regime-sensitive rebalancing materially improves outcomes.

These findings are quantified in Tables 5 and 6. Table 5 reports full-sample performance metrics for the U.S. portfolios. MinVar and Tangency portfolios both achieve an annualized Sharpe ratio of 0.261, outperforming the equal-weight portfolio (Sharpe 0.053), the standard 60/40 equity–bond benchmark (X60_40, Sharpe 0.043), and silver-only or SPX-only strategies. Notably, X60_40—implemented as a stylized 60% equity / 40% bond benchmark—delivers lower risk-adjusted performance and deeper maximum drawdowns than the gold-tilted MinVar allocations. Table 6 compares dynamic and static implementations: the RollingMinVar portfolio attains a Sharpe ratio of 0.994 with markedly smaller maximum drawdowns ($\text{MaxDD} \approx -0.24$) than StaticMinVar, equal-weight, or other static benchmarks. Figures 16–18 and Tables 5–6 together show that dynamic gold-tilted portfolios dominate traditional static allocations, including the canonical 60/40 benchmark.

5.4 U.S. Portfolio Weights, Crisis vs. Control, and Efficient Frontiers

Next, we examine how the optimizer allocates across assets over time. Figure 11 plots rolling 52-week minimum-variance portfolio weights in the United States. Gold’s weight increases sharply during major crises, while NASDAQ and SPX weights contract; bond exposure remains limited in the baseline optimization given their negative Sharpe ratios. Figure 12 applies an 8-week smoothing window to the same weights. The smoothed dynamics reveal clear regime shifts: in tranquil periods the optimizer tilts toward NASDAQ, but in stress periods it rotates aggressively into gold and away from equities.

Figures 13–15 extend this analysis to Europe, Japan, and the United Kingdom. In all three regions, gold’s share of the minimum-variance portfolio jumps during local and global

crises, whereas regional equity indices (STOXX 600, JP index, FTSE) and local bonds play a larger role in control periods. These regional results suggest that the “flight to gold” is not a U.S.-specific phenomenon but a robust feature of crisis-time portfolio construction.

Table 2 summarizes U.S. full-sample minimum-variance and tangency portfolio weights. The optimizer allocates roughly 25% to gold and 75% to NASDAQ in both the MinVar and tangency portfolios, with zero weight on silver, SPX, and both Treasury maturities. This is a striking result: over the full sample, silver is not selected at all in the mean–variance sense, and bonds are excluded due to their negative Sharpe ratios.

Crisis vs. control allocations and outcomes are detailed in Tables 3A and 3B. Table 3A shows that in crisis periods the MinVar portfolio holds approximately 74% gold and 26% NASDAQ, while in control periods it holds 100% NASDAQ and 0% gold or silver. Table 3B reports that GoldOnly and MinVar portfolios deliver positive or relatively stable performance in crises, whereas SPXOnly, X60_40, and equal-weight portfolios suffer large negative returns, high volatility, and severe maximum drawdowns. SilverOnly experiences high volatility and large drawdowns in both regimes, with only modest improvements in Sharpe ratios relative to equities. Together, Tables 2, 3A, and 3B show that gold is systematically selected and rewarded in optimization, while silver is consistently sidelined.

Figures 19–21 summarize these patterns visually. Figure 19 reports full-sample Sharpe ratios for the same set of portfolios, echoing Table 5: gold-tilted MinVar and tangency portfolios deliver the highest risk-adjusted performance, with silver-only strategies at the bottom of the distribution. Figure 20 compares Sharpe ratios in crisis vs. control periods and shows that crisis Sharpe ratios deteriorate strongly for equity-heavy and X60_40 portfolios, but remain relatively robust or positive for GoldOnly and MinVar allocations. Figure 21 displays annualized returns across the two regimes, with gold-heavy portfolios preserving or even improving returns in crises while equity-only and silver-only portfolios experience severe collapses.

Efficient frontiers and crisis-specific allocations complete the U.S. picture. Figure 22 reports optimal portfolio weights from weekly data, showing that gold plays a persistent role in high-Sharpe portfolios, particularly at lower risk levels. Figure 23 decomposes minimum-variance weights by crisis vs. control periods, reinforcing the sharp rotation into gold during stress. Figure 24 plots the U.S. efficient frontier in terms of annualized returns and volatility; the gold-tilted frontier dominates the 60/40 benchmark and equity-only points, indicating that investors can achieve higher Sharpe ratios and smaller drawdowns by substituting gold for part of the equity and bond allocation. Figure 25 zooms in on crisis-specific minimum-variance weights over time, documenting repeated spikes in gold weight during each major episode (dot-com bust, GFC, European debt crisis, COVID-19).

5.5 Regional Efficient Frontiers and Robustness Across Markets

Finally, Figures 26–30 and Table 4 evaluate whether these portfolio results are specific to the United States or generalize internationally. Figure 26 plots regional efficient frontiers based on median points for each region’s asset set. In all regions, frontier portfolios that include gold lie above and to the left of those that rely solely on local equities and bonds, indicating superior risk-return trade-offs. Figure 27 compares Sharpe ratios for dynamic versus static portfolios across regions; dynamic strategies that adjust allocations over time (following the U.S. RollingMinVar template) systematically outperform their static benchmarks, mirroring the robustness patterns in Table 6.

Figures 28–30 present regional efficient frontiers for Japan, the United Kingdom, and Europe, respectively, focusing on the trade-off between minimum variance and maximum return. In each case, the minimum-variance portfolios place substantial weight on gold and relatively little on silver or long-term bonds, while maximum-return portfolios tilt more heavily toward local equities and, in Europe and Japan, toward silver when the objective is pure return. This confirms that silver’s value is primarily in high-risk, high-return strategies rather than in risk-mitigating roles.

Table 4 provides the underlying regional weights used to construct these frontiers. For Europe, the minimum-variance portfolio puts roughly 50% in gold and the remainder in STOXX 600 and a small allocation to 10-year bonds, with zero weight on silver; the maximum-return portfolio, by contrast, places 100% on silver or on the equity index, depending on the region. In the United Kingdom, the minimum-variance portfolio uses a combination of gold and long-term gilts, while the max-return portfolio concentrates in FTSE. In Japan, minimum-variance portfolios again favor gold and the local equity index, with little or no role for silver or bonds. These regional results show that gold’s safe-haven and diversification properties are robust across currencies and markets, whereas silver consistently behaves like a high-beta, return-seeking asset rather than a defensive hedge.

6. Discussion and Interpretation

Taken together, the results paint a clear and internally consistent picture of what gold and silver actually do inside realistic portfolios. Across crises, asset classes, and regions, gold behaves much closer to a classic safe-haven asset than silver, but that safe-haven behavior is strongly state dependent and most valuable in the tails of the return distribution. In quiet markets, the optimizer largely ignores gold and loads into high-Sharpe equity exposure (especially NASDAQ in the U.S. sample). During severe stress, however, the model repeatedly rotates into gold-heavy minimum-variance portfolios, sharply reducing drawdowns and stabilizing cumulative wealth relative to equity-only, silver-only, and 60/40 benchmarks. Silver, by contrast, rarely appears in minimum-variance or tangency portfolios and delivers unstable, crisis-sensitive performance that is more akin to a speculative or high-beta equity substitute.

These patterns align well with the theoretical distinction between hedges and safe havens developed in the literature. On average, gold is not an “always-on” hedge that guarantees negative correlation with equities in every state; instead, it behaves more like a weak hedge in normal times and a strong safe haven in acute crises. This is exactly what we see in the rolling allocation and performance figures. In control periods, gold’s portfolio weight collapses toward zero and NASDAQ dominates the minimum-variance allocation. In crisis periods, gold’s weight surges, and the MinVar and gold-only strategies generate noticeably smaller drawdowns and better Sharpe ratios than equity-only and 60/40 portfolios. The empirical message is that gold is not an insurance policy you are paid to hold all the time, but it becomes extremely valuable when risk materializes in the specific form of large, correlated equity crashes and global “risk-off” shocks.

The results also clarify what gold does *not* protect investors from. Gold does not dominate equities on a long-run buy-and-hold basis; equities still deliver higher average returns over the full sample, and even gold-tilted frontier portfolios retain substantial equity exposure in order to achieve attractive Sharpe ratios. Nor does gold reliably hedge interest-rate risk: in our sample, long-term bonds earn negative Sharpe ratios, and gold’s role is primarily as a diversifier against equity drawdowns rather than as a substitute for duration. Similarly, gold does not fully insulate portfolios from macroeconomic or policy uncertainty that does not translate into sharp equity selloffs; its value is most obvious when equity markets are in deep distress and global yields are falling.

Silver’s role is almost the mirror image. The metal occasionally participates in equity rallies and, in some regional maximum-return portfolios, helps push the frontier outward on the high-risk, high-return end. But silver’s high volatility and tendency to sell off with equities in crises make it a poor candidate for risk reduction. The optimizer’s consistent zero or near-zero allocation to silver in minimum-variance and tangency portfolios—across countries and across crisis windows—is economically intuitive: when the objective is to minimize losses or maximize Sharpe ratios, silver’s incremental diversification benefits are more than offset by its instability and cyclical exposure. In that sense, silver behaves less like “poor man’s gold” and more like a leveraged pro-cyclical asset that belongs, if at all, in speculative rather than defensive allocations.

The superiority of dynamic, rolling portfolios over static benchmarks is another important dimension of the results. Static 60/40 and static equal-weight portfolios—despite their popularity in practice—suffer large drawdowns and low crisis Sharpe ratios in our sample. By re-estimating covariances and expected returns over rolling windows, the dynamic minimum-variance strategy implicitly learns when the environment is becoming more crisis-like and tilts into gold accordingly. This timing ability is visible in the rolling weight plots and is confirmed by the higher Sharpe ratios and smaller maximum drawdowns of the rolling strategies.

From a practical perspective, this suggests that gold's safe-haven value is best harvested through adaptive risk management rather than through a fixed "X% in gold at all times" rule.

Finally, the regional results strengthen the external validity of our conclusions. Efficient frontiers in Europe, Japan, and the United Kingdom tell much the same story as in the United States: gold systematically occupies a central position in minimum-variance portfolios, while silver is at best an occasional component of high-return, high-risk allocations. The fact that these patterns emerge across different equity indices, bond markets, and monetary regimes suggests that the safe-haven behavior of gold is not a purely U.S.-centric artifact. At the same time, cross-country differences in the exact magnitude and timing of gold's portfolio share remind us that safe-haven properties are not mechanical: they depend on local market structure, exchange-rate regimes, and the nature of each crisis.

In summary, the results show that gold protects investors primarily from *severe, equity-driven market crashes and associated global risk-off episodes*, and does so in a way that is robust across regions and portfolio constructions. Silver does not offer comparable protection and behaves more like a volatile, growth-sensitive asset than a reliable hedge. These findings reinforce and extend the existing safe-haven literature by demonstrating, within a unified mean–variance framework, that the economic value of gold as a safe haven is realized not through unconditional outperformance, but through targeted reduction of drawdowns and enhancement of risk-adjusted returns when they are needed most.

7. Conclusion

This paper set out to answer a focused question: *What exactly does gold protect investors from, and how does its performance compare with silver across crises, regions, and portfolio constructions?* Using weekly data on equities, long-term government bonds, and precious metals for the United States, Europe, Japan, and the United Kingdom, and embedding these assets in a mean–variance framework with both static and dynamic portfolios, we evaluated gold and silver not only in terms of correlations, but in terms of drawdowns, volatility, Sharpe ratios, and crisis-specific portfolio performance.

The evidence is clear that gold behaves much closer to a safe-haven asset than silver, though its protective properties are strongly state-dependent. In tranquil markets, gold contributes modestly to portfolio efficiency and is often largely excluded from minimum-variance allocations, which tilt toward high-Sharpe equity indices such as NASDAQ. In contrast, during major crisis episodes—the dot-com bust, the Global Financial Crisis, the European sovereign-debt crisis, and the COVID-19 shock—gold's portfolio weight rises sharply, especially in minimum-variance and rolling strategies. In these states, gold substantially reduces drawdowns and stabilizes cumulative wealth, outperforming traditional equity-only and 60/40 stock–bond benchmarks on a risk-adjusted basis. Silver, by comparison, behaves more like a high-beta asset: it occasionally boosts returns in risk-on environments but provides little

protection during deep equity selloffs and is rarely selected in optimal minimum-variance or tangency portfolios.

The portfolio results corroborate and extend the stylized facts in the safe-haven literature. Gold does not dominate equities on a long-run, unconditional basis, nor is it an “always-on” hedge against all sources of macroeconomic risk. Instead, its value is realized when risk materializes in the form that investors care about most: large, clustered equity losses and global “risk-off” shocks. In those states, gold-heavy minimum-variance and rolling portfolios deliver smaller maximum drawdowns and higher Sharpe ratios than static 60/40 and equal-weight portfolios, even though gold’s unconditional average return is lower than that of equities. Silver, in contrast, fails to replicate this pattern; it increases volatility, participates in equity drawdowns, and only occasionally contributes on the high-risk, high-return frontier.

A second key result is the superiority of dynamic, rolling strategies over static benchmarks. Portfolios that re-estimate covariances and expected returns over rolling windows and adjust weights accordingly—effectively “learning” when the environment is becoming more crisis-like—outperform static minimum-variance, 60/40, and equal-weight portfolios on both Sharpe ratio and maximum drawdown metrics. This suggests that the economic value of gold as a safe haven is best captured through adaptive risk management rather than through fixed, unconditional allocations. In practice, this means that long-horizon investors can enhance resilience by allowing gold’s portfolio weight to vary systematically with changing volatility and correlation regimes.

Finally, the regional evidence indicates that these conclusions are not unique to the United States. Across Europe, Japan, and the United Kingdom, efficient frontiers that include gold dominate those built only from local equities and bonds, and minimum-variance portfolios in each region assign substantial weight to gold while placing little or no weight on silver. Maximum-return portfolios, by contrast, tilt toward local equities and sometimes silver, reinforcing the idea that silver belongs, if at all, in speculative growth-oriented allocations rather than in defensive safe-haven roles. Overall, the cross-regional results strengthen the case that gold is a robust, though state-dependent, safe haven, while silver is better characterized as a cyclical, high-beta asset.

In sum, the analysis shows that gold protects investors primarily from *severe equity-driven crashes and global risk-off episodes*, reducing portfolio drawdowns and improving risk-adjusted performance when those protections matter most. It does not promise superior unconditional returns, nor does it consistently hedge all forms of macroeconomic or rate risk. Silver, meanwhile, does not provide comparable insurance and should not be treated as “poor man’s gold” in portfolios where downside protection is the primary objective.

8. Limitations and Directions for Future Research

The results in this paper should be interpreted in light of several limitations. First, the analysis is conducted within a standard mean–variance framework using weekly data. This approach is transparent and connects directly to portfolio choice, but it abstracts from non-normal return distributions, skewness, and tail dependence that might be better captured by downside-risk measures such as Value-at-Risk, Expected Shortfall, or drawdown-based optimization. It also assumes that investors care only about mean and variance, which may understate the value of assets that hedge rare disasters or extreme left-tail events.

Second, the crisis windows are defined *ex ante* using broad historical dates. While these windows align with major episodes identified in the literature, they inevitably smooth over differences between the onset, peak, and resolution phases of each crisis, and they treat “crisis” and “control” as binary categories. A more granular, model-based approach—such as regime-switching or volatility-state models—could provide a finer mapping between gold’s behavior and underlying risk regimes. Relatedly, the analysis focuses on a specific set of crises and a particular sample period; future research could investigate whether the conclusions hold in earlier eras (e.g., the 1970s inflation episodes) or in markets with different monetary and institutional structures.

Third, the implementation abstracts from transaction costs, taxes, and market frictions. Rolling minimum-variance portfolios implicitly involve turnover, particularly around crisis boundaries. Although the broad rotations we document—from equities into gold in stress and back into equities in calm periods—are relatively infrequent, incorporating realistic trading costs and implementation constraints would refine the assessment of how much of the theoretical performance advantage is realizable in practice. Similarly, we treat gold and silver as fully liquid, costless to trade, and continuously investable, which is an approximation for many institutional and retail investors.

Fourth, the asset universe is deliberately narrow: equities, long-term government bonds, gold, and silver, with a focus on developed markets. We do not consider other potential safe havens such as high-quality corporate credit, inflation-linked bonds, safe-haven currencies (e.g., the U.S. dollar or Swiss franc), or alternative assets like real estate, commodities, or Bitcoin. Including these additional candidates could sharpen the question of whether gold is a *unique* safe haven or simply one strong member of a broader class of defensive assets, and whether silver’s role changes once it competes with a wider set of cyclical and speculative assets.

These limitations point directly to several avenues for future research. One natural extension is to examine gold and silver in portfolio frameworks that explicitly account for downside risk and tail dependence, such as expected shortfall minimization, drawdown-constrained optimization, or regime-switching models calibrated to crisis vs. non-crisis states. Another is to broaden the asset universe and evaluate gold’s role relative to safe-haven currencies, inflation hedges, and newer digital assets, under realistic implementation

costs and constraints. A third direction is to study the microstructure and flow-based determinants of gold and silver prices during crises—for example, how ETF flows, central-bank purchases, or margin dynamics interact with safe-haven demand. Finally, there is scope to explore investor heterogeneity: different horizons, liability structures, and regulatory environments may place different values on the kind of state-dependent protection that gold provides.

Despite these caveats, the core message is robust: when evaluated through the lens of portfolios, rather than only correlations, gold emerges as a meaningful, state-dependent safe haven that protects investors from the worst equity-driven losses across multiple crises and regions. Silver does not. For practitioners, this suggests that the key decision is not whether to hold gold at all times, but how to integrate it into an adaptive risk-management framework that selectively leans on gold when it is most likely to justify its place in the portfolio.

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Appendix

Figure 1: United States: S&P 500 Total Annual Returns

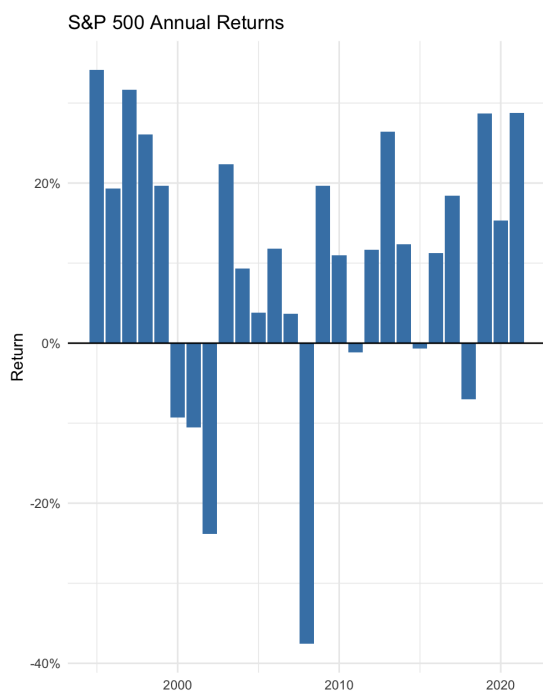


Figure 2: United States: SPX Drawdowns with Gold and Silver (Indexed)

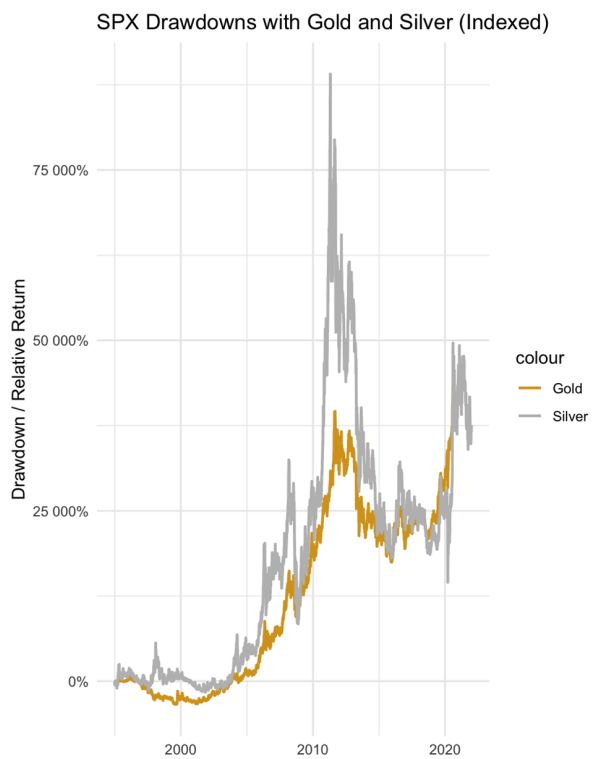


Figure 3: United States: Normalized Gold, Silver, Equities, and Treasury Yields (2000-2025)

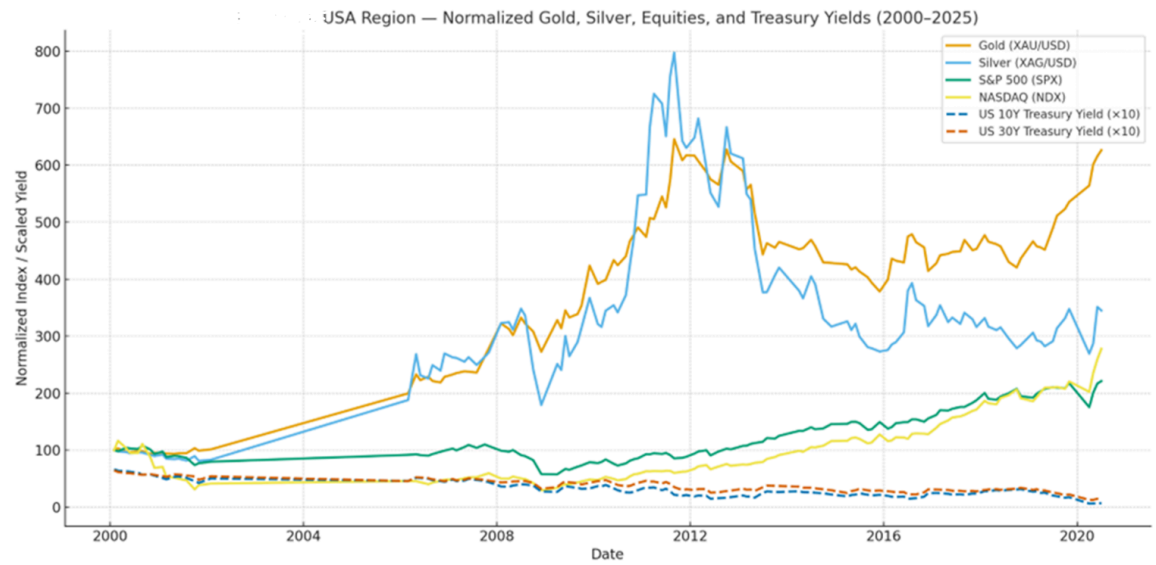


Figure 4: Global: 10-Year and 30-Year Treasury Yields- Across All Regions (1995-2025)

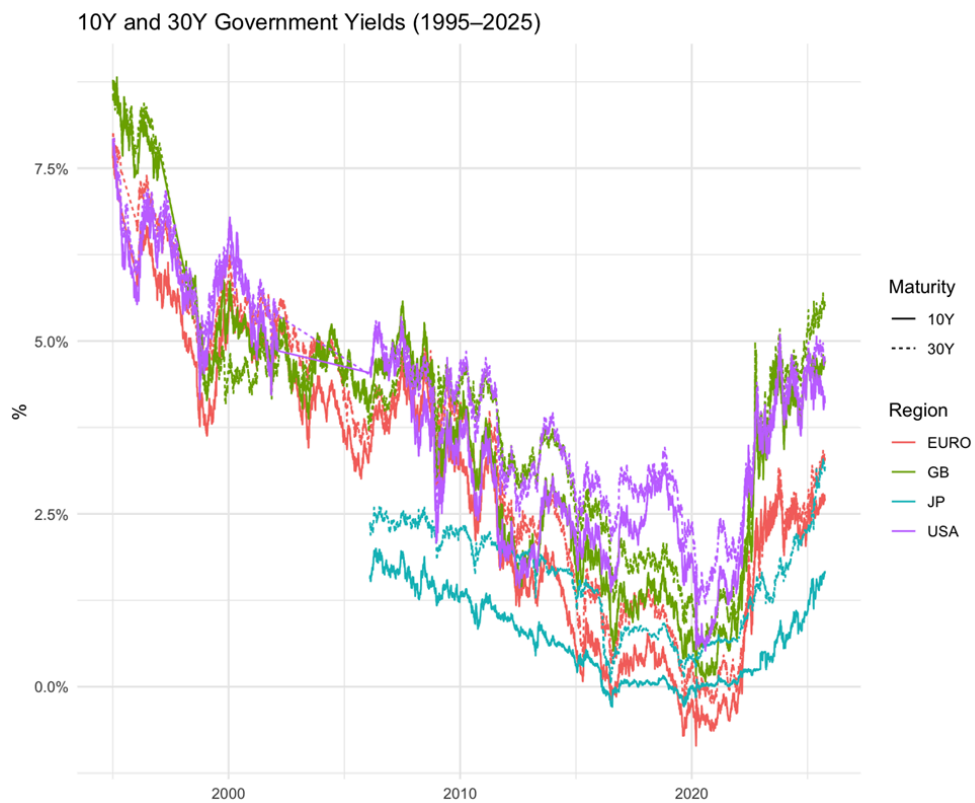


Figure 5: Europe and Japan: Gold and Silver in Europe and Japan (Normalized)

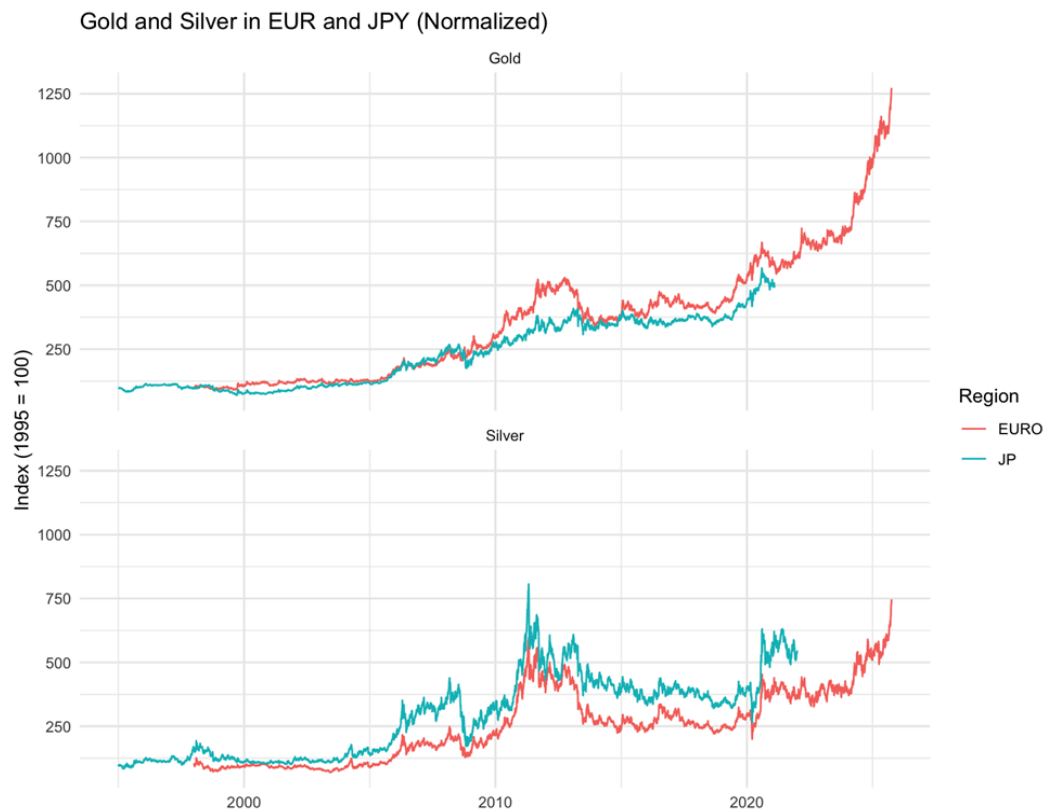


Figure 6: Global: Normalized Equity Index Levels (1995 = 100)

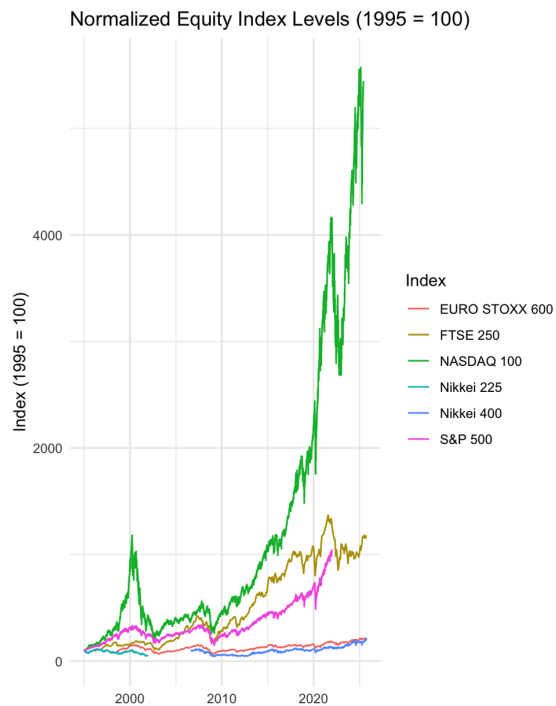


Figure 7: United States: Rolling 12-Month Drawdowns in S&P 500 with metal overlays

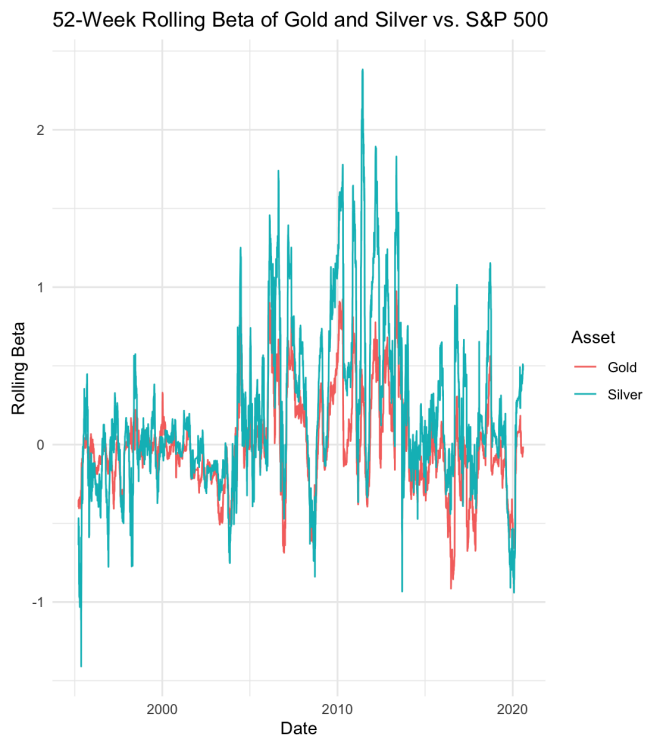


Figure 8: United States: Gold & Silver Volatility vs. S&P 500 Volatility

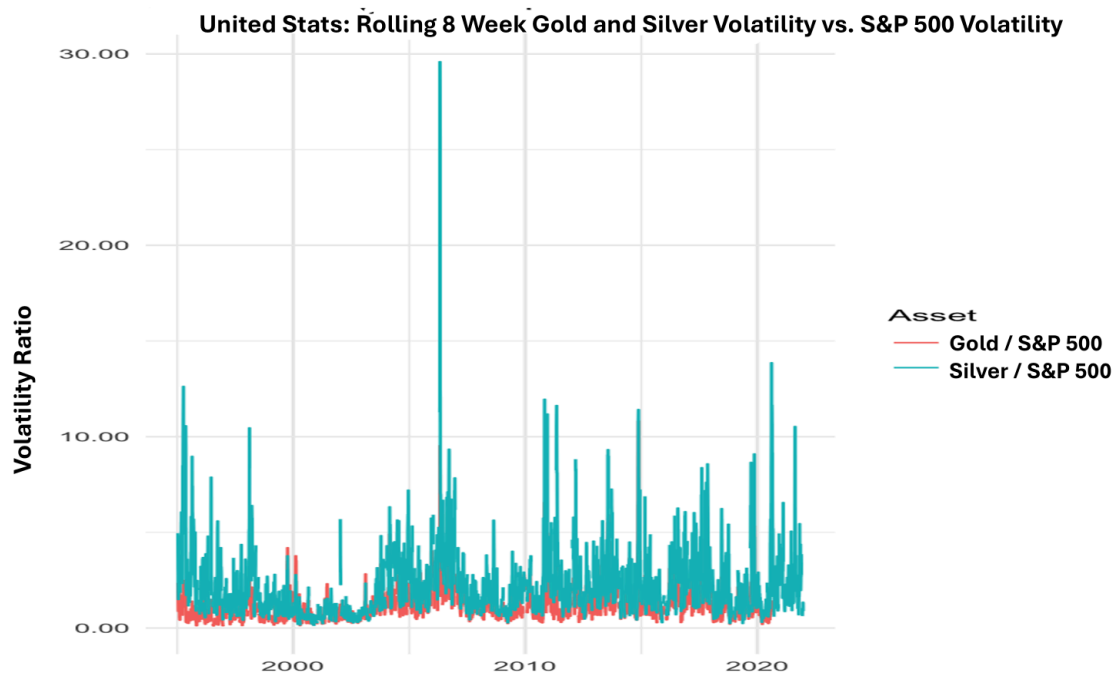


Figure 9: Normalized Gold Prices by Region (1995-2025)

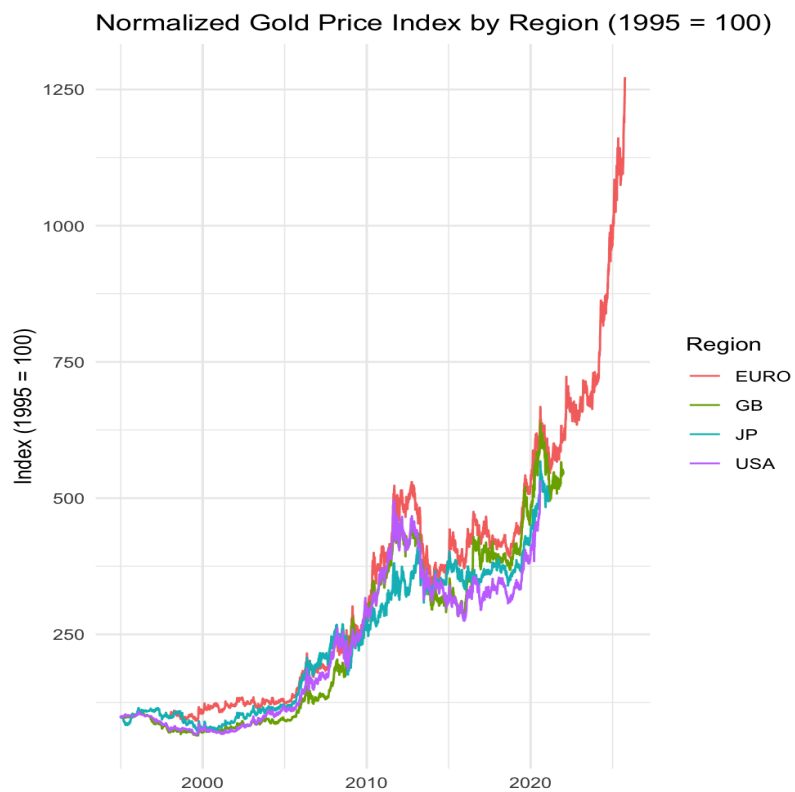


Figure 10: Normalized Silver Prices by Region (1995-2025)

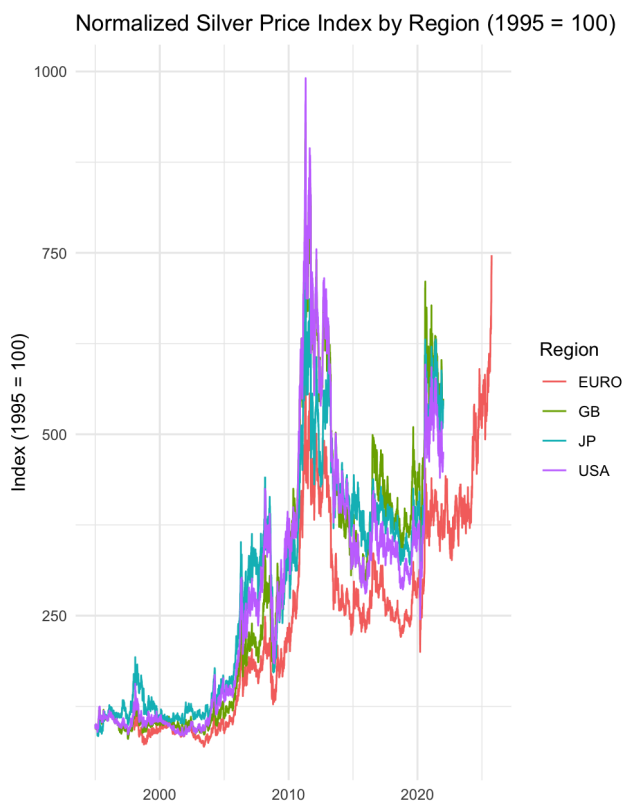


Figure 11: United States Rolling 52-Week Minimum-Variance Portfolio Weights (1995 - 2020)

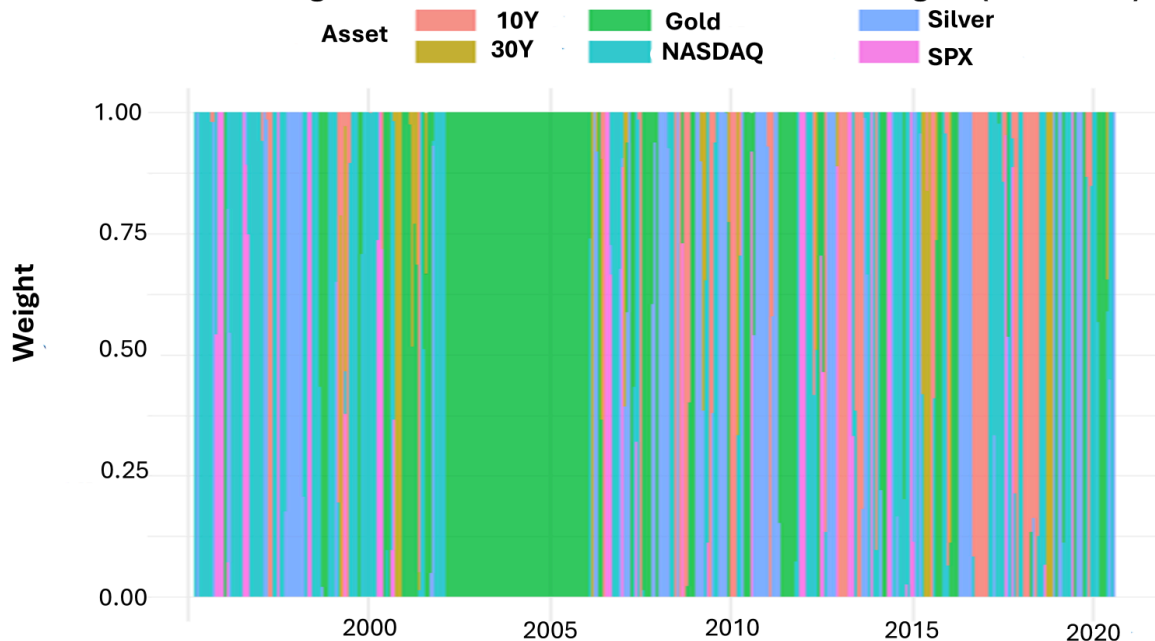
United States: Rolling 52-Week Minimum Variance Portfolio Weights (1995-2020)

Figure 12: United States Rolling 8-Week Smoothed Minimum-Variance Portfolio Weights (1995 - 2020)

Rolling 8-Week Smoothed Minimum-Variance Portfolio Weights (1995–2025) : United States

Crisis periods shaded | Smoothed from 52-week rolling optimizations

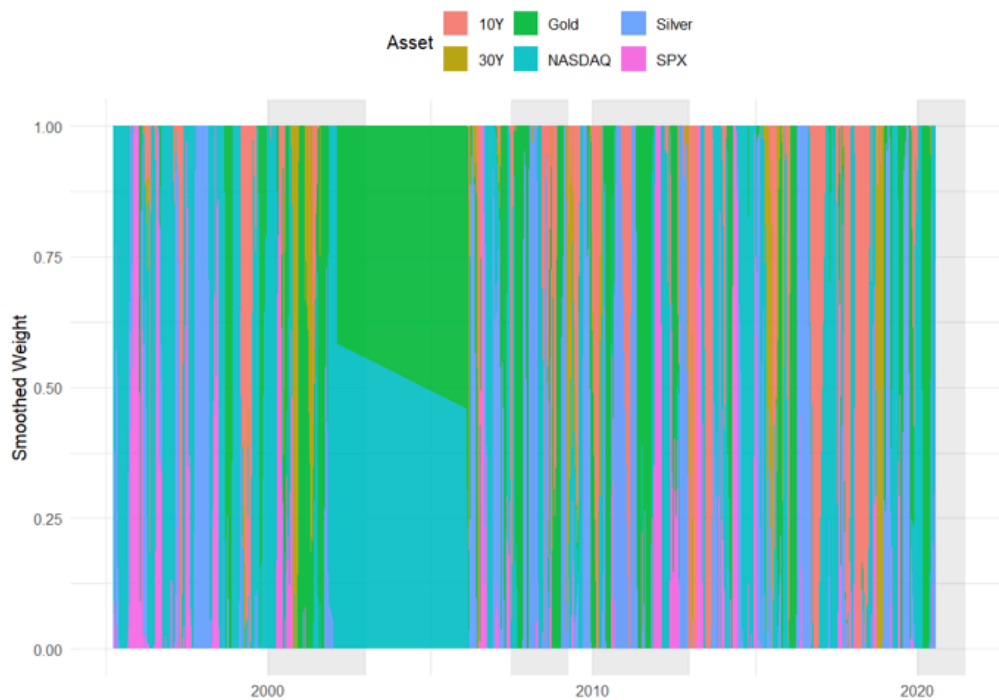


Figure 13: Europe Rolling 8-Week Smoothed Minimum-Variance Portfolio Weights (2000-2025)

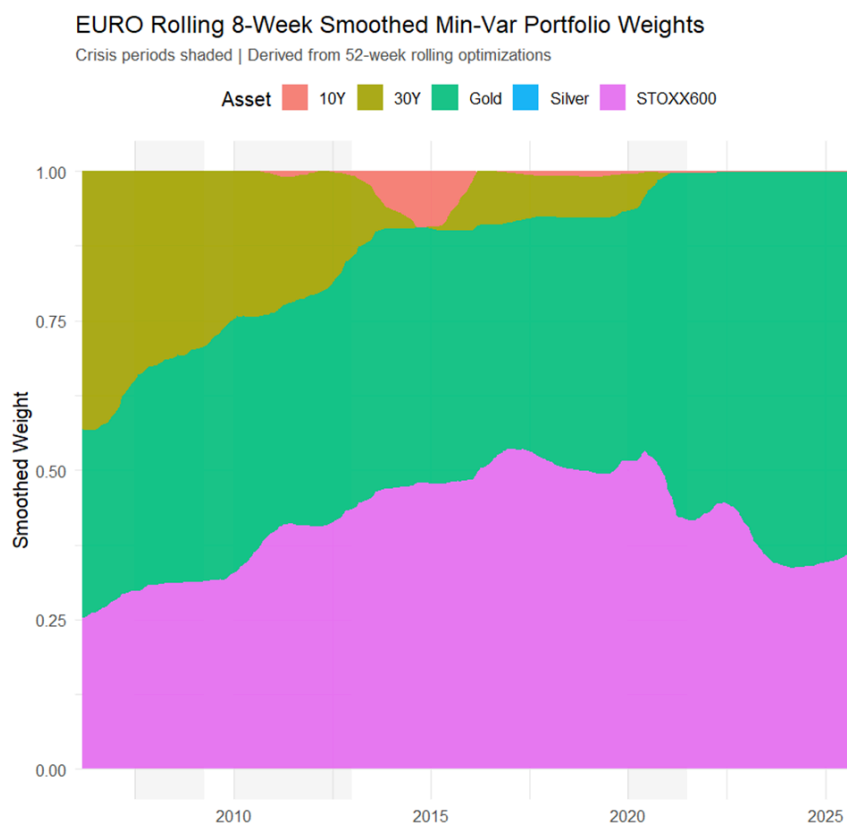


Figure 14: Japan Rolling 8-Week Smoothed Minimum-Variance Portfolio Weights (2000-2025)



Figure 15: United Kingdom Rolling 8-Week Smoothed Minimum-Variance Portfolio Weights (2004-2025)

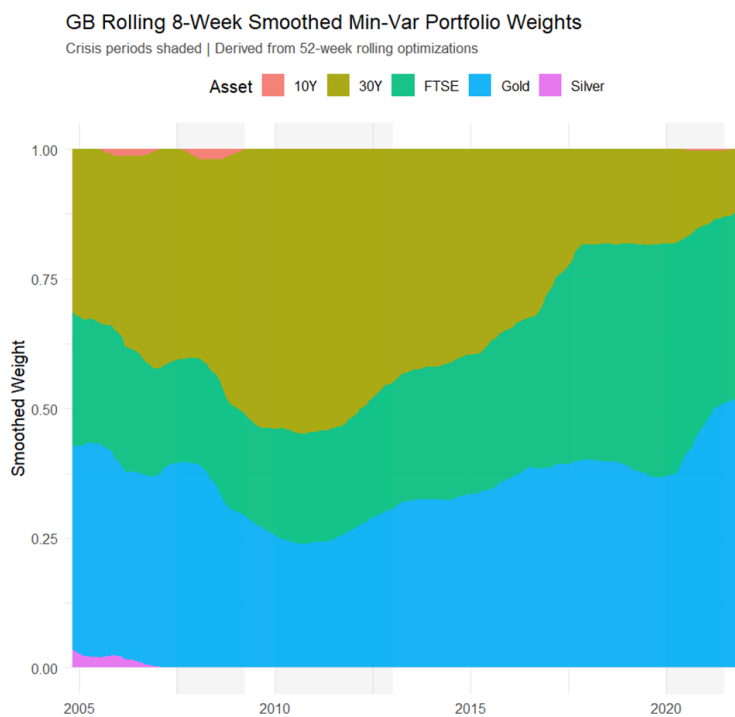


Figure 16: United States Cumulative Wealth of Key Portfolios

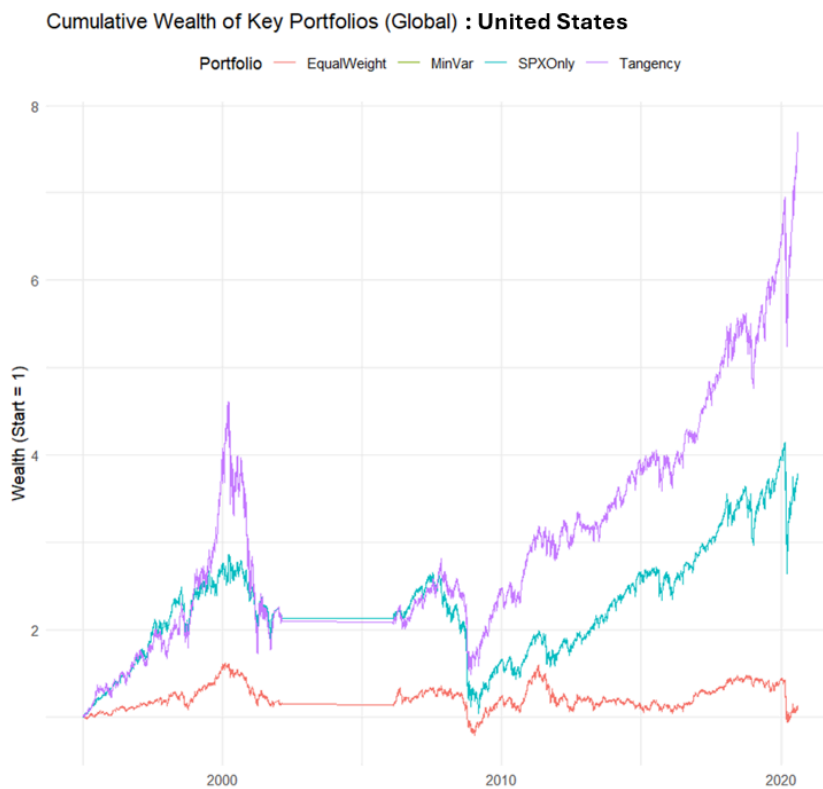


Figure 17: United States Drawdowns: Minimum-Variance vs S&P 500

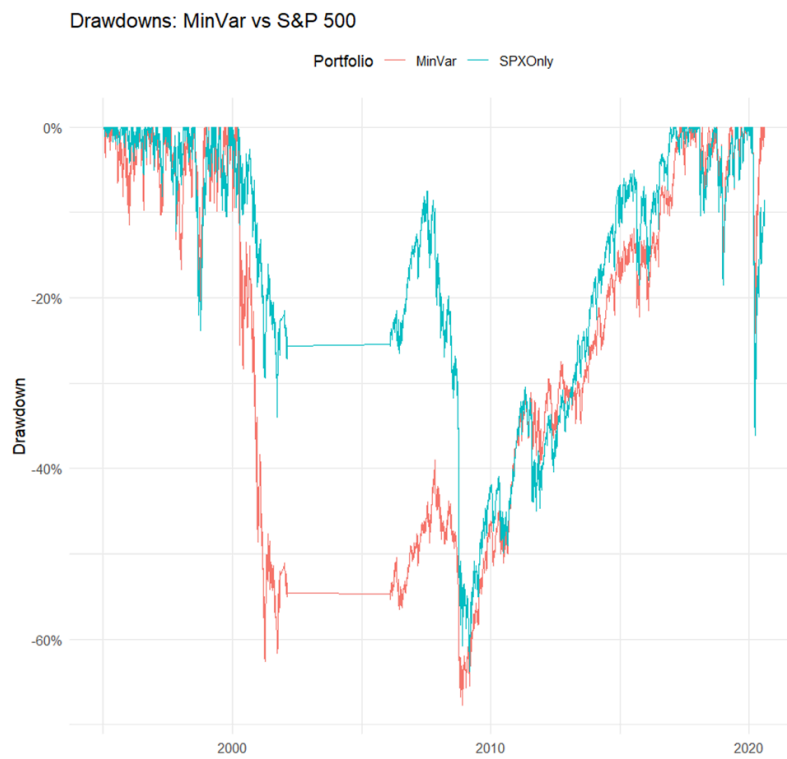


Figure 18: United States Rolling vs Static Portfolio Wealth Paths

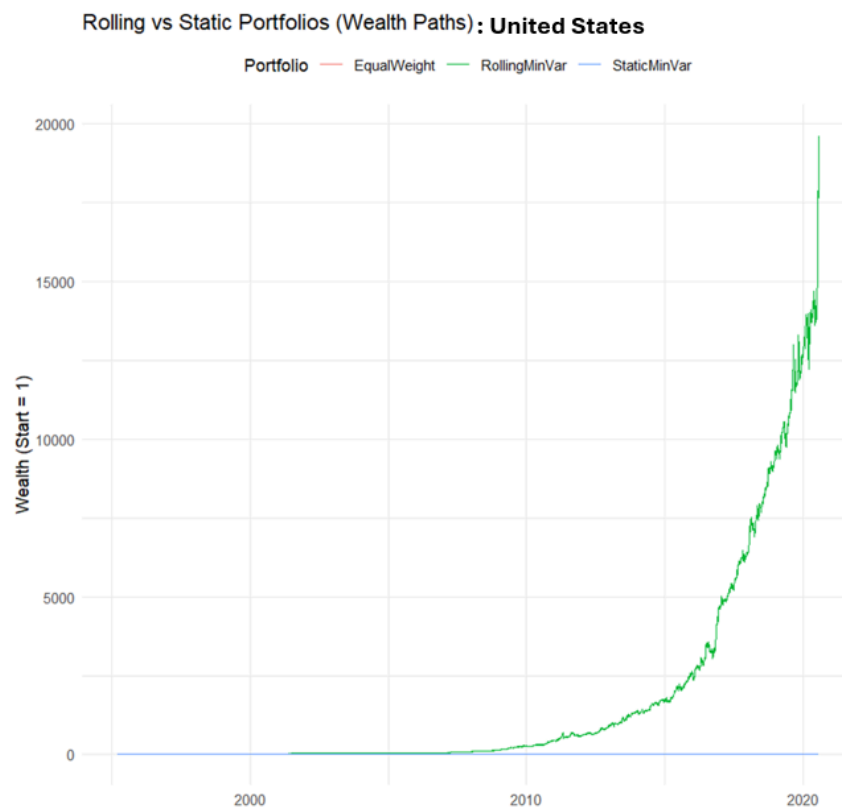


Figure 19: United States: Portfolio- Full Sample Sharpe Ratios

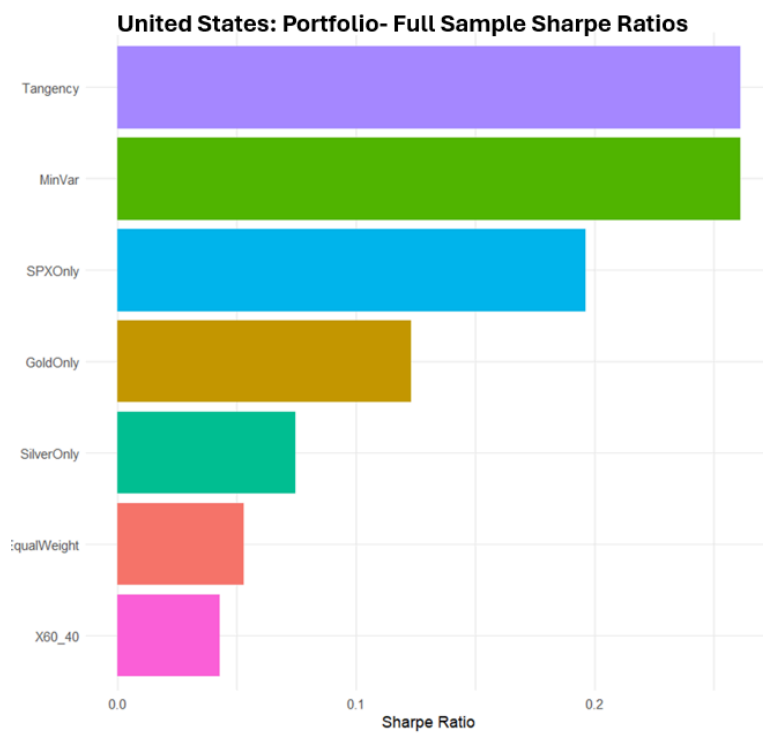


Figure 20: United States: Sharpe Ratios- Crisis vs. Control

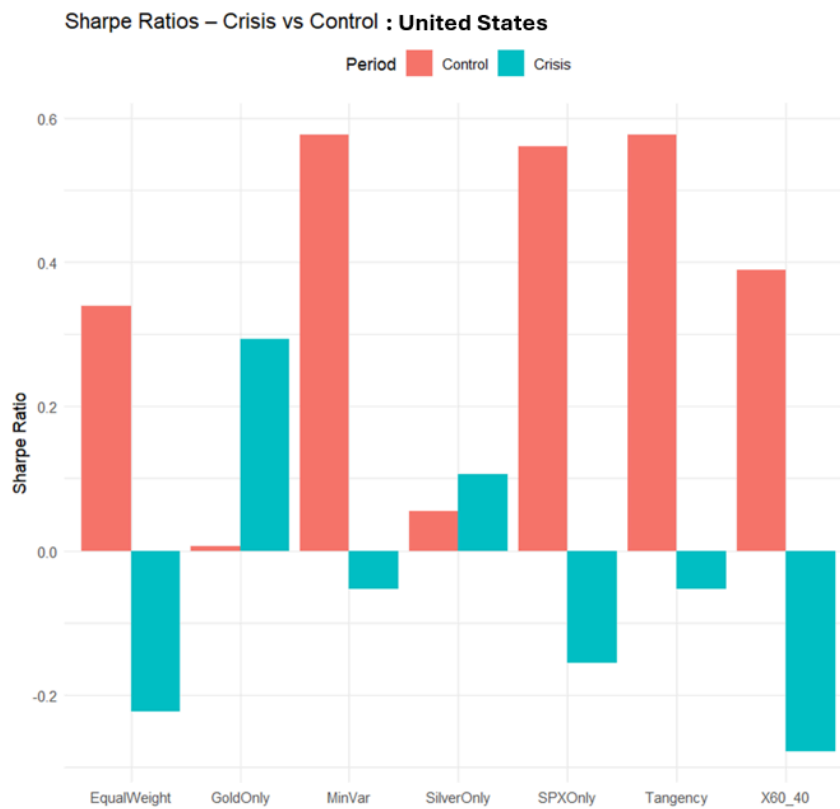


Figure 21: United States: Annualized Returns- Crisis vs. Control



Figure 22: United States: Optimal Portfolio Weights (Weekly Data)

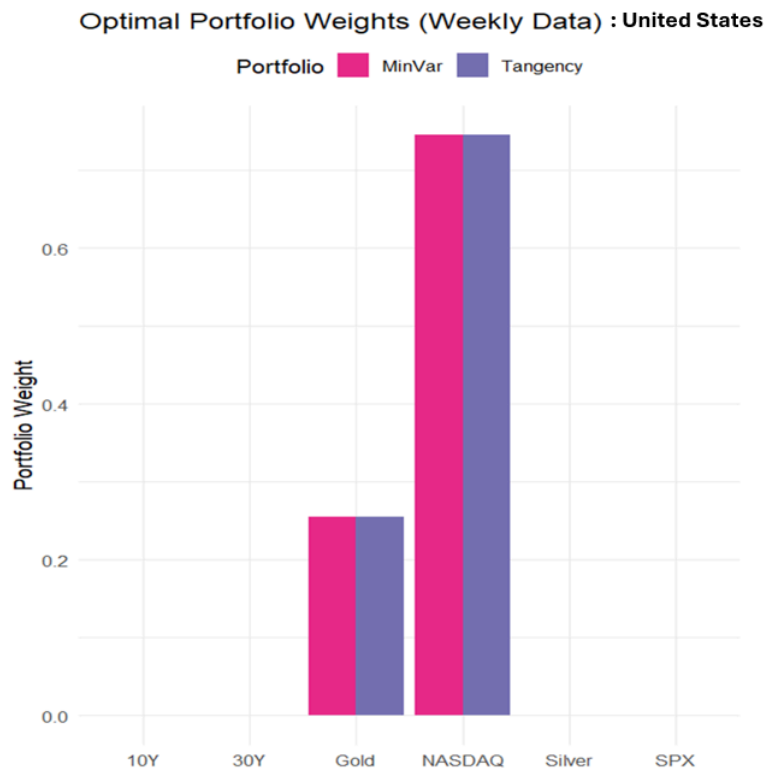


Figure 23: United States: Minimum Portfolio Weights: Crisis vs Control

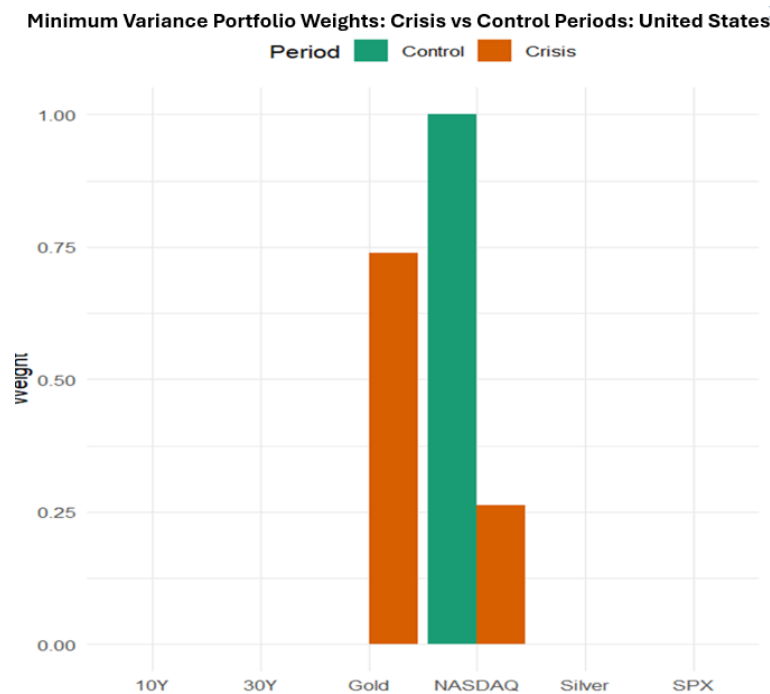


Figure 24: United States: Efficient Frontier- Weekly Returns, Annualized Percentages (1995-2025)

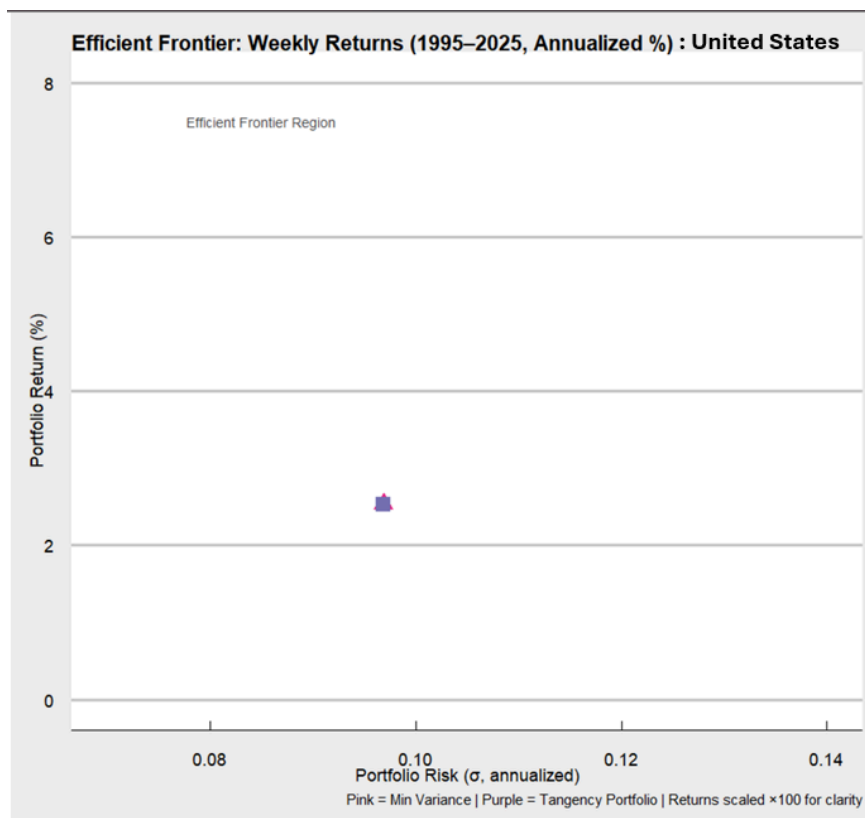


Figure 25: United States: Crisis-Specific: Minimum Variance Portfolio Weights (Weekly data)

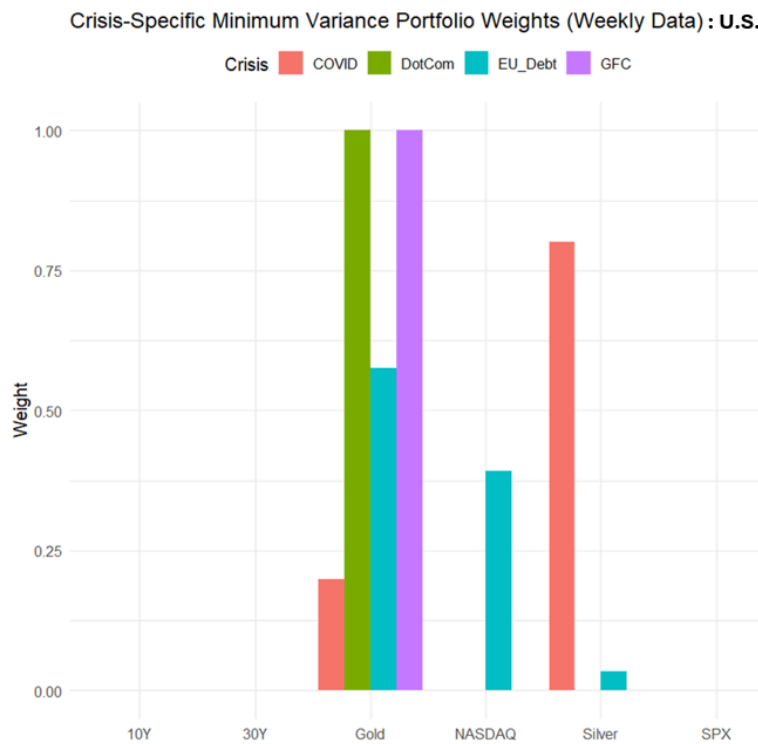


Figure 26: Regional Efficient Frontier- Median Points

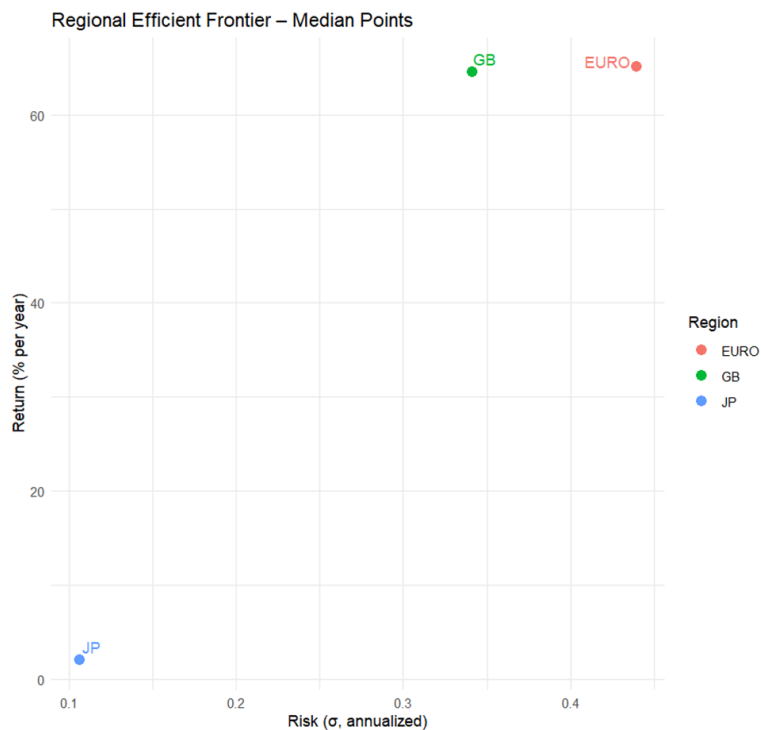


Figure 27: Regional- Dynamic vs Static Portfolios- Sharpe Comparison

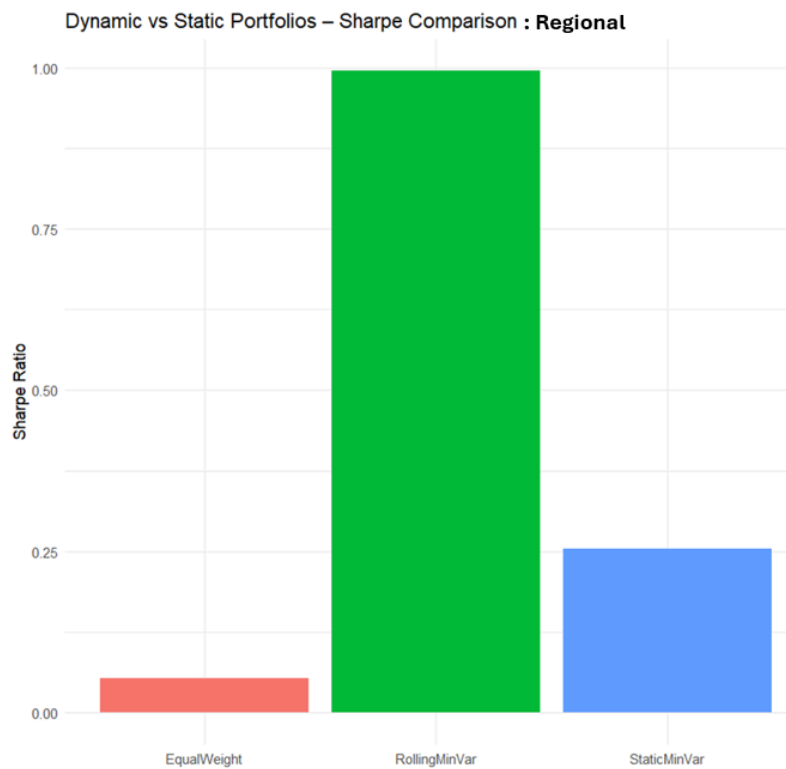


Figure 28: Japan: Regional Frontier- Minimum Variance on Max Returns, as percentages

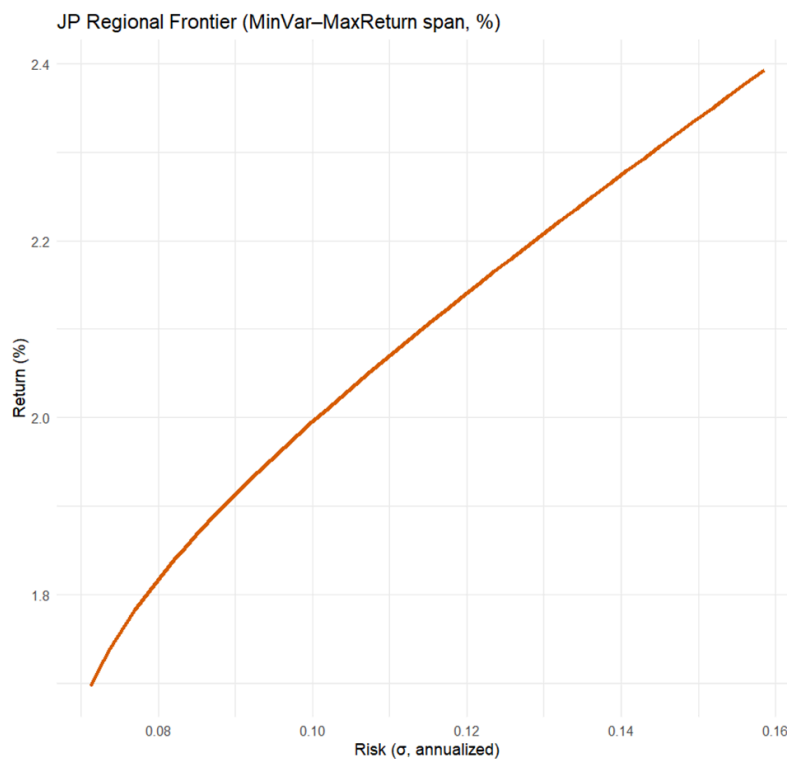


Figure 29: United Kingdom: Regional Frontier- Minimum Variance on Max Returns, as percentages

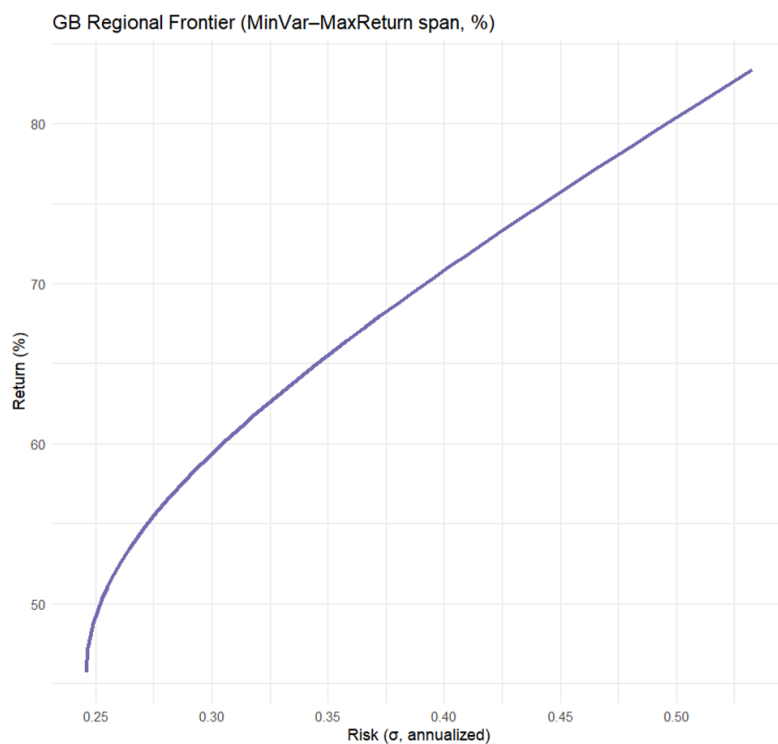
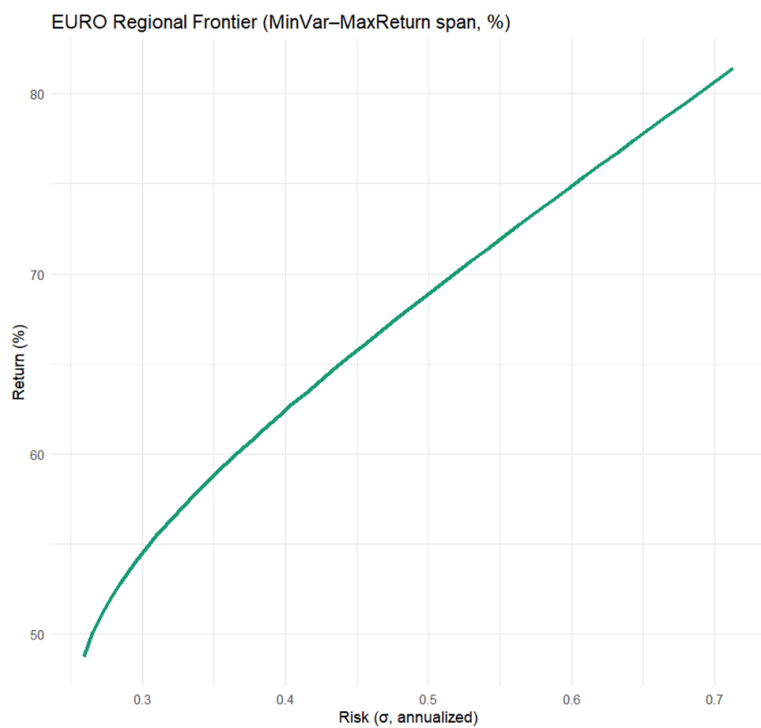


Figure 30: Europe: Regional Frontier- Minimum Variance on Max Returns, as percentages



Empirical Portfolio Analysis and Asset Performance Metric Tables

Table 1: U.S. Summary Statistics

Asset	Region	Ann_Return	Ann_Vol	Sharpe
Gold	U.S.	0.92%	7.50%	0.128
Silver	U.S.	0.99%	13.31%	0.077
SPX	U.S.	1.74%	8.86%	0.200
NASDAQ	U.S.	3.08%	12.77%	0.244
10Y	U.S.	-2.68%	16.20%	-0.164
30Y	U.S.	-1.74%	11.74%	-0.145

Table 2: U.S. Full-sample MVP & Tangency Weights

Asset	Portfolio	Weight
Gold	MinVar	0.25488
Gold	Tangency	0.25488
Silver	MinVar	0
Silver	Tangency	0
SPX	MinVar	0
SPX	Tangency	0
NASDAQ	MinVar	0.74512
NASDAQ	Tangency	0.74512
10Y	MinVar	0
10Y	Tangency	0
30Y	MinVar	0
30Y	Tangency	0

Table 3A – U.S. Crisis vs Control Weights

Asset	Period	Weight
Gold	Crisis	0.73824
Gold	Control	0
Silver	Crisis	0
Silver	Control	0
SPX	Crisis	0
SPX	Control	0
NASDAQ	Crisis	0.26176
NASDAQ	Control	1
10Y	Crisis	0
10Y	Control	0
30Y	Crisis	0
30Y	Control	0

Table 3B – U.S. Crisis vs Control Performance

Portfolio	Period	Ann_Return_pct	Ann_Vol_pct	Sharpe	MaxDD_pct	Ann_Return	Ann_Vol	MaxDD
EqualWeight	Control	176.29%	518.96%	0.340	-1612.62%	0.02	0.05	-0.16
EqualWeight	Crisis	-227.03%	1020.39%	-0.222	-6807.41%	-0.02	0.10	-0.68
GoldOnly	Control	3.96%	660.28%	0.006	-4388.09%	0.00	0.07	-0.44
GoldOnly	Crisis	263.05%	897.31%	0.293	-2526.90%	0.03	0.09	-0.25
MinVar	Control	420.07%	728.16%	0.577	-2225.36%	0.04	0.07	-0.22
MinVar	Crisis	-69.17%	1311.91%	-0.053	-7257.10%	-0.01	0.13	-0.73
SPXOnly	Control	362.56%	646.64%	0.561	-2387.39%	0.04	0.06	-0.24
SPXOnly	Crisis	-189.87%	1220.63%	-0.156	-7240.95%	-0.02	0.12	-0.72
SilverOnly	Control	65.11%	1189.20%	0.055	-6132.50%	0.01	0.12	-0.61
SilverOnly	Crisis	165.54%	1569.20%	0.105	-6823.67%	0.02	0.16	-0.68
Tangency	Control	420.07%	728.16%	0.577	-2225.36%	0.04	0.07	-0.22
Tangency	Crisis	-69.17%	1311.91%	-0.053	-7257.10%	-0.01	0.13	-0.73
X60_40	Control	220.07%	564.51%	0.390	-2048.09%	0.02	0.06	-0.20
X60_40	Crisis	-322.40%	1155.83%	-0.279	-7677.51%	-0.03	0.12	-0.77

Table 4 – Regional Weights

Region	Asset	MinVar	MaxReturn
EURO	Gold	0.50335	0
EURO	Silver	0	1
EURO	STOXX600	0.49448	0
EURO	10Y	0.00217	0
EURO	30Y	0	0
Region	Asset	MinVar	MaxReturn
GB	Gold	0.47881	0
GB	Silver	0	0
GB	FTSE	0.32899	1
GB	10Y	0	0
GB	30Y	0.1922	0
Region	Asset	MinVar	MaxReturn
JP	Gold	0.58508	0
JP	Silver	0	1
JP	JP_Index	0.37365	0
JP	30Y	0.04127	0

Table 5 – U.S. Performance Metrics (Full Sample)

Portfolio	Ann_Return_pct	Ann_Vol_pct	Sharpe	MaxDD_pct	Ann_Return	Ann_Vol	MaxDD
MinVar	253.12%	968.18%	0.261	-6775.84%	0.03	0.10	-0.68
Tangency	253.12%	968.18%	0.261	-6775.84%	0.03	0.10	-0.68
EqualWeight	38.67%	730.25%	0.053	-5155.83%	0.00	0.07	-0.52
X60_40	34.96%	816.63%	0.043	-5770.91%	0.00	0.08	-0.58
GoldOnly	92.37%	749.74%	0.123	-4915.31%	0.01	0.07	-0.49
SilverOnly	99.38%	1330.98%	0.075	-8149.45%	0.01	0.13	-0.81
SPXOnly	174.05%	885.99%	0.196	-6371.80%	0.02	0.09	-0.64

Table 6 – U.S. Robustness / Dynamic vs Static

Portfolio	Ann_Return_pct	Ann_Vol_pct	Sharpe	MaxDD_pct	Ann_Return	Ann_Vol	MaxDD
RollingMinVar	1067.35%	1073.43%	0.994	-2432.04%	0.11	0.11	-0.24
StaticMinVar	247.79%	972.09%	0.255	-6775.84%	0.02	0.10	-0.68
EqualWeight	39.56%	733.17%	0.054	-5155.83%	0.00	0.07	-0.52